Machine Learning-Based Breast Cancer Diagnosis Using Support Vector Machines and Neural Networks

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***Abstract*—In this paper we compare the results obtained with Support Vector Machines (SVMs) and Neural Networks (NNs) for breast cancer diagnosis of the Wisconsin Diagnostic Breast Cancer dataset. After preprocessing and hyperparameter tuning, we have obtained an SVM model with accuracy of 98.25% and low computational cost. We have also obtained an optimized NN model with 100% accuracy and good cross-validation results. The results of our project showcase that both svm and neural approaches can achieve high diagnostic accuracy when they are optimized, with each offering different trade-offs between efficiency and their predictive performance in the medical classification tasks.**

***Index Terms*—Breast cancer diagnosis, machine learning, support vector machine, neural networks**

I. Introduction

Breast cancer is one of the most life-threatening and unfortunately common disease worldwide. To improve patient survival, making an early and accurate diagnosis is essential. By automating classifications of tumors, machine learning has become more and more crucial I medical diagnostics by using biological and imaging derived data. Since it contains detailed cellular features retrieved from fine-needle aspiration samples, the Wisconsin Diagnostic Breast Cancer (WDBC) dataset is a well known benchmark for evaluating such predictive models. The complexity of the correlated medical features make the selection of a reliable classification model crucial in order to balance predictive power and generalizations.

With models spanning from basic machine learning techniques to sophisticated deep learning systems, recent research has shown great advancements in breast cancer classification using the WDBC dataset. A number of recent contributions are summarized in Table 1.

Issues of overfitting, data leaking and clinical relevance of recent research of high accuracy on breast cancer classification are addressed by our project through a transparent and carefully controlled comparison between Support Vector Machines and Neural Networks on the Wisconsin Diagnostic Breast Cancer dataset to evaluate and compare predictions of both methods with a focus on strong generalization and genuine clinical relevance.

II. Background

A. Dataset description

The dataset of our project, the Breast Cancer Wisconsin (Diagnostic) dataset is based upon digitized images of fine needle aspirate (FNA) breast cancer mass. Every sample is composed of a single cell nucleus that is extracted from a medical image. The problem we attempt to tackle is to identify if a tumor is malignant or benign. This decision is based upon many quantitative features of the cell nuclei itself.

Now moving onto the details of the dataset; it contains a total of 569 samples, none have any missing values. It consists of thirty features that describe the properties inside the cell nuclei; including an ID and binary diagnosis (either malignant or benign). They are computed from by a form of digitized images of fine needle aspirate of breast mass. These numerical features include the characteristics themselves of the cell nuclei: as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. For every feature, it has three additional features including the mean, standard error, the worst.

It must be noted that the class distribution is imbalanced; including 357 benign cases versus 212 malignant. Though, this dataset has a very clean nature, making it a known candidate for binary classification. It has high features that are informative with clinical importance. This dataset was collected during research on robust linear discrimination and is publicly available for use through the UCI Machine Learning

TABLE I
RECENT RESEARCH ON BREAST CANCER CLASSIFICATION USING THE WDBC DATASET

Study	Methodology	Accuracy	Key Contribution
Essa [1]	CNN with Gaussian feature engineering	100%	Combined feature transformation with deep learning for perfect classification performance
Al Reshan [2]	Stacking ensemble with neural meta-learner	99.89%	Developed advanced ensemble architecture with strong robustness and sensitivity
Al Mamun [3]	Optimized SVM and Logistic Regression	99.12%	Showed that careful hyperparameter tuning can rival more complex systems
Cheng & Yu [4]	Logistic Regression framework	98.25%	Focused on interpretability, reproducibility, and clinical transparency
Alam [5]	CWV-BANN-SVM hybrid ensemble	100%	Confidence-weighted hybrid system integrating ANN and multiple SVM classifiers

Repository. Hence it is often the candidate for supervised learning in any medical diagnosis.

B. Support Vector Machines (SVM) Background

Firstly, we will tackle what Support Vector Machines actually are. SVMs are a type of supervised learning algorithm that are used for classification. To put it simply, SVMs find a separating boundary between classes. However, the idea behind it is to find the optimal separating hyperplane that maximizes the margin between the two classes. This margin is the distance between the decision boundary itself and the nearest data points to it called support vectors.

The decision boundary is the hyperplane in the form of a line equation:

$$\mathbf{w}^T\mathbf{x} - b = 0$$

; where w is a vector that defines the orientation of the line and b is the bias (meaning the offset of this line). The classification is decided based upon which side of the decision boundary the data lies. That SVM boundary is chosen such that those two margin lines distance is maximized. This basically converts it into an optimization problem: find the best separating boundary while keeping that boundary as simple as possible. The "weight vector" controls the position and direction of the that separating hyperplane. Minimizing its norm basically means making that boundary as stable and wide as possible all while classifying the training data correctly.

However, in real world messy datasets; classes are obviously not linearly separable. This moves us to use non-linear decision boundaries as the solution. This is achieved through kernel function, often called the kernel trick. What it does is it implicitly transforms our data into higher-dimensional spaces in which it would actually be linearly separable. In our project, we use the radial bias function (RBF). It allows us to have flexible decision boundaries that would be able to capture the complexity of the breast cancer dataset. In the project, RBF is used to model the non-linear relationship between the tumor features and their diagnosis.

When properly regularized, SVM is effective in high dimensional spaces. In addition, its reliance on its support vectors results in a computational efficient model since the whole data is not taken into account.

C. Neural Networks Background

Now this type of model is actually inspired by our very own biological brain. It is made of a system of connected processing units called artificial neurons, quite like our brain. Organized in layers, including: the input layer, one or more of the hidden layers, and the output layer.

The way it works is a three step process inside a neuron: each of these neurons read input signals (such as the features like radius, texture, area), then it multiplies each input's by an importance weight and adds them up (the bigger the weight the more importance and vice versa, the bias adjusts the sensitivity), then after computing a weighted sum of those inputs, the neuron applies a non-linear activation function (such as sigmoid) to transform the output into a bounded value between 0 and 1, allowing the model to represent complex, non-linear relationships rather than simple linear ones. The ReLU formula can also be used as an activation function to output 0 if the input is negative, stays the same if positive; helping the network learn faster and focus on meaningful signals.

The end goal simply put, is to learn complex mappings from our input features to output labels all by adjusting the weights of the connections between the neurons. It does not rely on manually designed features, it's capable of automatically learning those hierarchal feature representations. Each of the hidden layers transforms the given input into more abstract representations, this helps the model to capture our complex non linear shape of the data.

Now for the training part, the way it happens is it involves minimizing the cost function. The cost functions is what measures the difference between the predicted outputs and the actual label. This is done using the gradient descent (an iterative optimization algorithm used to find the local minimum of a differentiable function) combined with back propagation. Backpropagation is the process where the neural network checks how wrong its prediction was, then works its way backward (hence the name back propagation) through the network to figure out which weights contributed most to that error and adjusts them to improve future predictions.

In our project we used the multilayer perceptron (MLP). Our data moves through its multiple layers to learn patterns that are inside the tumor features. Using labeled examples of benign and malignant tumors, we trained the model as it kept adjusting the weights internally to improve its prediction accuracy. In order to avoid overfitting, we used regularization techniques (adding a penalty term (alpha) to discourage overly large weight values) and early stopping (by stopping training once the performance stopped improving such that the risk of excessive fitting to noise is minimized) which helps the model perform well on data it hasn't seen before. Neural Networks are well suited for the breast cancer dataset as they help capture the non linear interactions between the medical features.

III. Methodology

A. Support Vector Machines (SVM) Methodology

The supervised learning Support Vector Machine model was developed in order to classify breast tumors as either malignant or benign. Firstly the data set was preprocessed. We started by removing unnecessary features, namely the patient ID and an empty unnamed column as they do not provide any numerical data for our model. Then the categorical diagnosis labels were then converted into numerical form using the label encoding so that benign and malignant cases could be processed mathematically by our model.

Then, in order to improve efficiency as well as reducing redundancy in our model, highly correlated features have been identified through correlation analysis. Features with correlation values that were above 0.90 were examined, and one feature from each highly correlated pair was removed based on lower variance, ensuring that more informative variables were kept. This process reduced dimensionality while preserving predictive strength and helped prevent unnecessary complexity.

The next step after the dataset was preprocessed is to split our data into training, validation, and testing subsets. We used a stratified split, meaning ensuring to preserve the original class distribution so that the model does not become biased. Sixty percent of the data was used for training, twenty percent for validation, and twenty percent for final testing. Then feature scaling was applied in order to improve SVM performance using the standard scaler. This standardized all numerical inputs, which is crucial for SVM as it relies on distance.

Our SVM model used the Radial Basis Function (RBF). The RBF kernel trick will help us capture the breast cancer's non linear nature. The RBF kernel allows the model to capture more complex decision boundaries. The hyperparameter tuning that took place was conducted by a manual grid-style search experiment across multiple C regularization values and the gamma kernel coefficient. The C parameter controlled the balance between maximizing the margin size but also minimizing the classification error. While the gamma controls how flexible the decision boundary will end up being. Then finally, the validation accuracy determines the optimal combination of our C and gamma values.

After the search has took place and the hyperparameters were decided, the training and validation sets were combined to create the final training dataset. Then, the SVM model's performance was evaluated on the unseen data using accuracy, F1-score, confusion matrix analysis, precision-recall curves, and learning curves. The computational complexity was also measured based upon analyzing the number of support vectors, which in turn is what directly affects the prediction's efficiency. In order to examine the most important features, permutation importance analysis was used to identify the most important features that have higher influence.

This method ensured that our SVM model has been optimized to reduce redundancy and overfitting, as well as computational efficiency, and stressing on reducing false negatives as they are most important in medical datasets.

B. Neural Networks Methodology

Now we move onto the neural networks approach in which we use a Multilayer Perceptron (MLP); a feedforward artificial neural network. It is made to model relationships that are complex and non linear like our dataset with its tumor characteristics and diagnosis outcomes. Similarly to the SVM pipeline, the dataset was loaded and normalized using the standard scaler, this ensures stable training behavior. Firstly, data exploration took place through histograms and correlation matrices to better understand feature relationships. Since neural networks are able to handle correlated features effectively, all major features were kept.

The first neural network had a single hidden layer of a 100 neurons since it is a balanced starting point. It has both enough learning capacity but also avoids any unnecessary complexity. Afterwards, each neuron processed the weight inputs then applies the activation function, and finally passes that information forward through the network. In order to minimize prediction error, the model was trained using supervised learning in which the internal weights were updated as backpropagation took place.

Cross validation was used in order to assess the generalization performance. The first model showed high accuracy yet in order to make sure that the model was not overfitted, a second model was created in which we enabled early stopping. By halting the training once performance stopped improving; early stopping helped prevent overfitting to reduce excessive memorization of the data. Stratified splitting took place in order to make sure there was balanced class distribution when represented. When early stopping took place; although it improved generalization, it reduced the performance and introduced more false negatives which is fatal in medical cases, it was necessary to optimize.

In order to effectively find the best model configuration, the GridSearchCV function was used to test the many combinations in the neural network. Multiple architectures were tested including shallow and deeper ones ranging from one hidden layer of 50 or 100 neurons to two hidden layers like (50,100). And then, two activation function were compared: the ReLU and tanh. In our project, tanh performed better at capturing the medical dataset. Then we tackled the learning rate at which we control how quickly the model updates its weights as well as regularization strength alpha which prevents overfitting was optimized to allow flexibility but also controlling complexity.

The final optimized neural network used a two-layer architecture (50 and 100 neurons), tanh activation, early stopping, a 0.01 learning rate, and low regularization. This captures the complex feature relationships in our medical dataset but also maintaining strong generalization. The model achieved perfect test performance and high cross-validation accuracy, showcasing both excellent predictive power and minimal overfitting.

IV. Results and Analysis

Now this section aims to interpret our results and precise numbers obtained from our SVM and Neural Network models on the Breast Cancer dataset using a required criteria of performance, accuracy, F1-score, computational complexity, and training and testing time.

A. Support Vector Machine (SVM) Results

In order to capture the complexity of the breast cancer dataset, the SVM model used a Radial Basis Function (RBF) kernel, after tuning the hyperparameters across multiple values of regularization strength (C) and kernel flexibility (gamma). The model achieved:

- Accuracy: 98.25%
- F1-score: 97.56%
- Training Time: 0.0159 seconds
- Testing Time: 0.0009 seconds
- Computational Complexity: 41 support vectors

These results show that the SVM model obtained excellent classification but also kept low computational cost. The learning curve in figure 1 shows that when small training samples are introduced, the F1-score begins at 1.00, meaning the model of course can memorize small data. But as more training samples were used the training score now stabilizes at around 0.98. Meanwhile, the cross-validation score starts out at a lower rate of about 0.90 and rises when more data is added, finally settling near 0.97. The two curves converging shows that generalization is well and that the model is not severely overfitted. To simply put it, as the model receives more exposure to the data, it becomes more accurate on new data and it performs consistently.

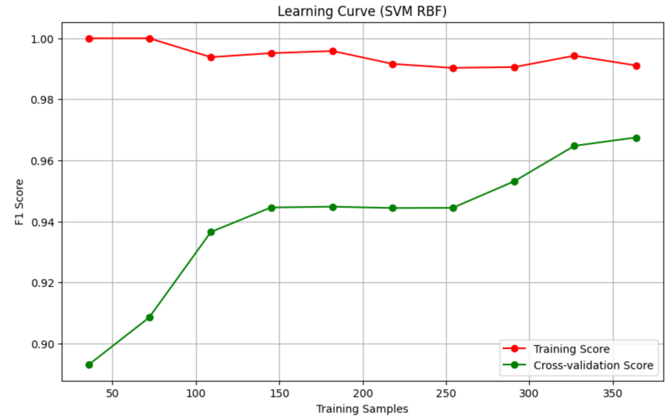

Fig. 1. Learning Curve of the SVM model

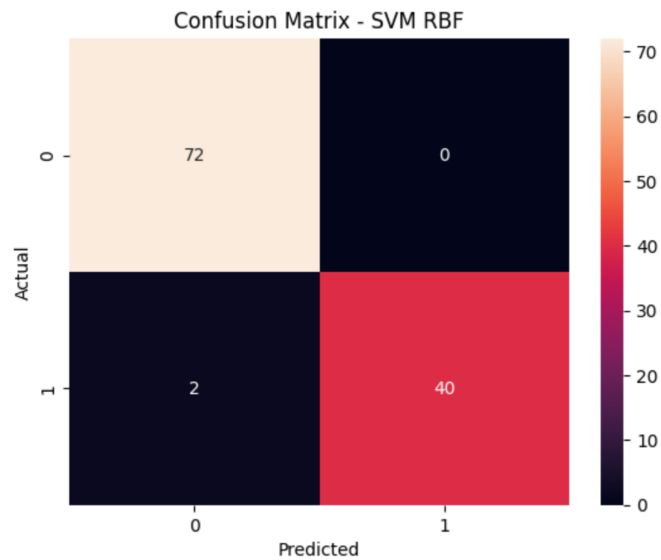

Fig. 2. SVM Confusion Matrix

The confusion matrix also shows more insight into the SVMs performance and diagnostic safety. As seen in the confusion matrix in figure 2; the model has correctly classified 72 malignant tumors (true positives), 40 benign tumors (true negatives), 0 false positives, 2 false negatives. This is important in medical cases since false negatives, meaning missing a malignant tumor, is most fatal. With only two false negatives and no false positives, the model show cases strong clinical reliability.

Another graph to evaluate and validate the effectiveness of the model is the precision-recall. As seen in figure 3; the curve maintained almost perfect precision for the most of the recall values, with an average precision (AP) of 0.99. In other words, when the model predicts the tumour as a malignant one, it is almost always correct. The slight dip on precision only occurs as the recall reaches its absolute maximum, and that is to be expected when forcing the model to capture almost every positive. In general, the SVM has a very good trade-off between predictive performance, speed and computational complexity.

B. Neural Networks Results

The Neural Network model that was initially used a standard Multilayer Perceptron (MLP) with one hidden layer of a 100 neurons. This original model has achieved perfect test accuracy of 100%, an F1-score of 100%, training time that is 0.4084 seconds, testing time that is 0.0030 seconds, and a cross-validation accuracy of 99.56%. Although this was highly impressive, it raised concerns about possible overfitting. To investigate that issue, an early stopping version was created. As seen in figure 4, both of the models began with similar cost values, but the early stopping model has halted much earlier before fully reaching the minimum cost achieved by the original model. This shows that early stopping has prevented

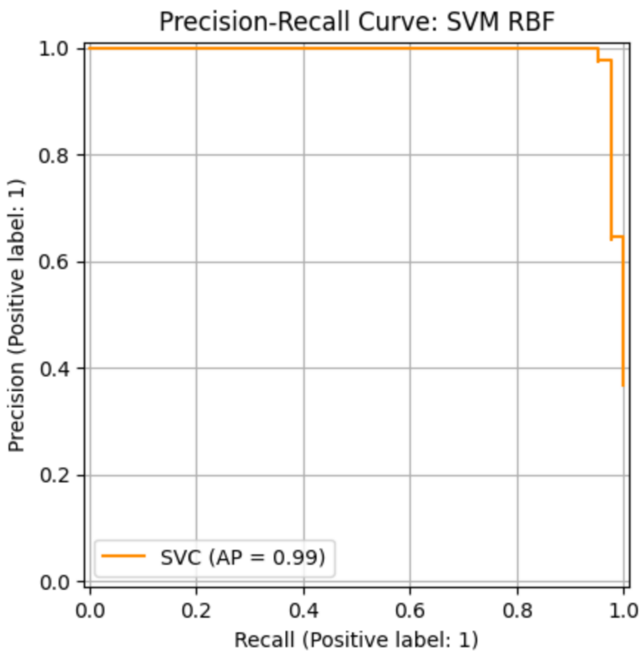

Fig. 3. Precision-recall SVM

the model from fully optimizing itself, and that possibly reducing its predictive power in exchange for stricter overfitting prevention. On the other hand, the original model has continued to learn until the cost nearly approached zero, while the early stopping model halted too early. When evaluated, the early stopping model produced an accuracy of 92% , an F1-score of 88%, and had increased its false negatives rates. This huge performance decline shows that early stopping may have been overly restrictive for the model.

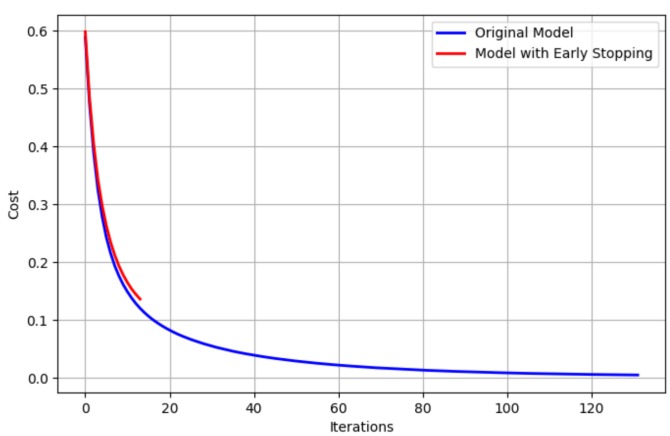

Fig. 4. Early Stopping Graph

Now in order to improve performance while preserving generalization, GridSearchCV was used to optimize the hidden layer architecture, the activation function, the learning rate, regularization factor (alpha). The search has found that the best optimized configuration inluded:

- Hidden layers: (50,100)
- Activation: tanh
- Learning rate: 0.01
- Alpha: 0.0001
- Early stopping enabled

This optimized model has achieved:

- Accuracy: 100%
- F1-score: 100%
- Training Time: 1.0374 seconds
- Testing Time: 0.0058 seconds
- Cross-validation Accuracy: 98.68%

The confusion matrix confirms this perfect classification as seen in figure 5. Now this shows near perfect performance on the test set but also keeping strong cross-validation results, showing that the optimized neural network has successfully balanced the complexity and the generalization.

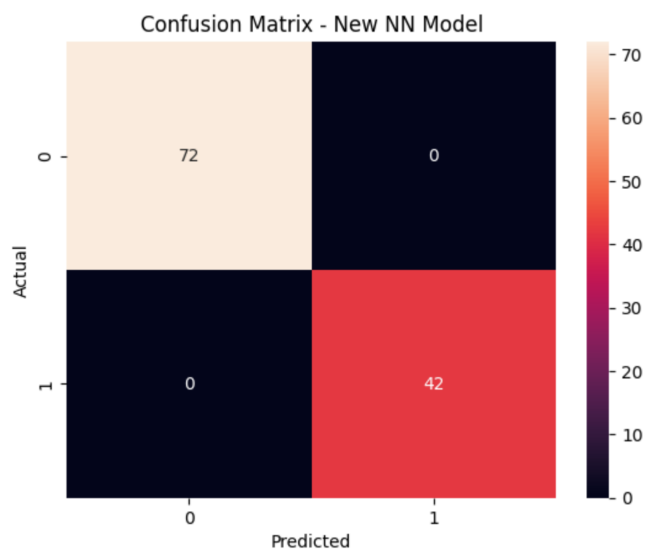

Fig. 5. NN Confusion Matrix

C. Comparative Performance and Literature Analysis

As seen in table 2, the results show that the two models tested are on par with the recent literature methods including more complex deep learning or ensemble models. The SVM model provides good accuracy with a low cost. This makes it suitable for efficient application. In addition, the neural network model shows perfect performance. It has proper tuning and generalizes well. This demonstrates that proper data preparation and optimization enables more simple, interpretable models to perform as well as more complex models.

V. Conclusion

This project has showcased that Support Vector Machines and Neural Networks can be used for classification of breast cancer using the Wisconsin Diagnostic dataset. The SVM model has a low computational cost and is quite robust. We

TABLE II
ACCURACY COMPARISON BETWEEN PROPOSED MODELS AND RECENT LITERATURE

Model	**Accuracy**	**Recent Literature Comparison**
SVM (RBF)	98.25%	Optimized SVM + Logistic Regression, 99.12% [3]
Optimized Neural Network	100%	CNN + Feature Engineering, 100% [1]

could see that the Neural Network model has achieved extremely high classification performance and strong generalization performance when it was carefully tuned and optimised. The results of our project are comparable to the stated models in recent literature. In the medical field, it is important to have models that are fast and simple with low false negatives rate, and SVM shows that models can be simple and still provide good performance for medical applications. In the future, it might be interesting to test this models on more different datasets. In addition, increasing the interpretability of these models could be helpful. Finally, it might be worth trying and seeing if these models can contribute to the reduction of false negatives even more in SVM, which are the most important factor in the medical field.

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